Student's Name……

WAIRAKA MODERN SS-JINJA

S.3 END OF TERM 1, 2026

Uganda Certificate of Education

MATHEMATICS

TIME: 2 hours 15 minutes

INSTRUCTIONS:

*This paper of **two** PARTS; **I** and **II.** It has **five** examination items.*

*PART I has **three** compulsory items.*

*PART II has **two** Items**.** Answer **one** item from this part.*

*Answer **four** examination items in all.*

*Any additional item(s) answered will **not** be scored.*

Graph paper is provided.

Silent, non-programmable scientific calculators and mathematical tables with a list of formula may be used.

PART I

Answer all items from this part

ITEM ONE

Mukyawe and Gladys bought an acre of land in Jinja district at Shs 7,000,000 on 01/06/2025 where they contributed 4,000,000 and 3,000,000 respectively. They portioned it into 8 plots and sold each plot at Shs. 1,500,000. On each plot, the dealer (A person who brings the buyer) is given Shs. 90,000, the LC1 chairperson is given 10% of the selling price and each witness is given 20,000. They only allow five witnesses to sign on the land agreement. Mukyawe and Gladys shared 7% of the actual excess money after selling all the plots in the ratio of their money contribution. They agreed to buy land again on the day they will both be off duty. Mr. Mukyawe is always off duty after every 4 days and Madam Gladys is off duty after every 7 days.

Task;

(a) How much did Mukyawe and Gladys get from the excess money?

(b) How much money will they use to buy land on the day they will meet again?

(c) When will they buy the land?

ITEM TWO

Julius and Nadia, a S.3 Students owns bookstore received an order to package 200 books for a customer. Previously, Julius package 48 books using 2 meters of packing tape and 168 books using 7 meters of packing tape. Both Julius and Nadia plan to start eggs and tea business in the school so they decided to go to Kiteezi eggs and nutritional center.

Julius bought 4 trays of eggs and 3kg of sugar and paid Ugx 58000. Nadia had Ugx 40000 and she bought 2 trays of eggs and 4kg of Sugar and she remained with only Ugx 1000. The friends of these students failed to know the prices of a tray of eggs and that of a kilogram of sugar since the two students where not given the receipts.

Task:

a) (i) Establish a mathematical relationship between the number of books package and the amount of packing tape used.

(ii) Determine the amount of packing tape required to package 200 books.

b) Help their friends to determine the price of a tray of eggs and a kilogram of sugar

ITEM THREE

At a given hardware shop, the manager requested the owner to increase on the number of casual workers who help to load cement for clients since the customers had now increased. The owner requested the manager to present record of the sales so as to determine the mean sales per day. The record of the numbers of bags of cement sold that day to 40 clients was as below

49	**40**	**45**	**36**	**67**	**32**	**45**	**42**
40	**42**	**61**	**48**	**45**	**67**	**51**	**53**
35	**46**	**50**	**43**	**47**	**37**	**47**	**47**
49	**30**	**62**	**48**	**53**	**41**	**53**	**43**
45	**49**	**51**	**44**	**48**	**38**	**49**	**48**

The owner needs this information presented graphically to make a decision. If 75% of the clients bought more than 51 bags of cement**,** a new worker will be brought to the shop.

Task:

a) Represent the information on a frequency distribution table with the first class being 30 – 34 to help determine the mean.

b) i) Represent the above information on a statistical graph which can be used to determine if a new worker is to be brought to the hardware shop.

ii) Will the new worker be brought to the hardware shop? Give a reason

PART II

Answer only one item in this part

ITEM FOUR

A parent intends to make shopping of scholastic materials for his children who are going back to school for a new term by the names of Jane, Mary and Darin. They budgeted as below basing on the list of requirements that they were given by their class teachers.

- Jane: 6 exercise, 3 pencils, 2 Graph books, 3 pens

- Mary: 3 pencils, 1 Graph book, 6 exercise books, 3 pens
- Darine: 2 Graph books, 4 exercise books, 3 pencils and 5 pens

At that time the prices were 1 Graph book Shs 2000, 1 pencil Shs 100, 1 exercise book Shs 1,500 and 1 pen Shs 500.

On reaching school they found out that the canteen manager had increased prices of the items by 10% and also they found out that the school administration had decided that on each item listed they should increase the number for each by 2 since school administration had decided that the students extend by two weeks when the term ends in order to compensate for the time students had lost the previous term.

Before leaving their home, they were given by their father Shs 200,800 so that they can finish the clearing process at school and then after wards they share equally the remaining money to be used as their pocket money.

Task:

a). Assuming they were to buy the items before going to school, using the matrices help the father to determine how much he would give to each child.

b). By use of matrices determine how much each child paid to the canteen attendant in order to acquire the items.

c). Help the children determine how much each shared as pocket money after buying the items from the school canteen.

ITEM FIVE

A local youth group in your district wants to install a solar-powered security light at a triangular plot of land in their community.

The length of each of the sides of a plot of land is 60m.

Their plan is to construct a circular fence around the plot of land and ensure that the solar powered security light is situated at the center of the fence.

The local youth group also wants to determine the area covered by the fence and use it to determine the amount of space, that extends beyond the plot of land, but which is within the fence.

The extra land beyond the triangular plot of land will be sold later and $1m^2$ of land is worth Ugx 50,000.

The group has approached for yourhelp.

Task.

As a learner of Mathematics, help the local youth group clearly;

(a) Construct the circular fence and locate the position of the security light from each of the comers of the plot of land.

(b) Find the area covered by the fence and the area of space that extends beyond the plot, within the fence, where necessary give your response to one decimal place.

(c) Determine the amount of money they will get after selling the extra piece of land.

WINNERS NEVER QUIT

THE END